Tema | Dia | Mes | Año | ESTILO

HW05 - Object Oriented Modeling and UML

Summary of slides

About : 1.05.01 , 1.05.07 , 1.05.08 , 1.05.09 and 1.05.10.

1.05.01

Conceptual Modeling

- Any time during the software development process. From a variety of perspectives:
  - Analysis
  - Design
  - Specification
  - Implementation
  - Deployment

UML (Unified Modeling Language), including:

- Use case diagrams
- Class diagrams
- Interaction diagrams
- State charts

For design software system in layers:

- Each layers should well-defined abstraction
- Each layer may use components in own layer

Layers:

Uses Interface, logic, persistent, communications, operation system

1.05.07

UML is:

expressive
semi-formal
capable of supporting incremental development
process independent
Customizable.

UML is and customizable object-oriented modeling language whose origins date back to the evolution of object-oriented programming from the 1960s to the 1990s.

Notas

Tema | Día | Mes | Año | ESTILO

Building Blocks

It consists of modeling elements (structural, behavioral, organizational, and annotational) that act like words to form diagrams (sentences) and models (books)

Diagrams

Types of Diagrams

Diagrams are broadly categorized into structural diagrams (such as class, object, component, and deployment diagrams) and behavioral diagrams (such as use case, sequence, collaboration, state chart, and activity diagrams)

1.05.08

Actors

Entities outside the system that interact with it, such as users, other software systems, or hardware devices.

They represent roles and can be organized through generalization.

Use cases

Ideas representing what users need to accomplish (goals) and capturing the user-visible functionality of a system without focusing on design choices

Relationships

- Extends «extend»
- Includes «include»
- Generalization

Use cases capture functional requirements, define system.

Notas

Tema | Día | Mes | Año | ESTILO

Building Blocks

It consists of modeling elements (structural, behavioral, organizational, and annotational) that act like words to form diagrams (sentences) and models (books)

Diagrams

Types of Diagrams

Diagrams are broadly categorized into structural diagrams (such as class, object, component, and deployment diagrams) and behavioral diagrams (such as use case, sequence, collaboration, state chart, and activity diagrams)

1.05.08

Actors

Entities outside the system that interact with it, such as users, other software systems, or hardware devices

They represent roles and can be organized through generalization.

Use cases

Ideas representing what users need to accomplish (goals) and capturing the user-visible functionality of a system without focusing on design choices

Relationships

- Extends «extend»
- Includes «include»
- Generalization

Use cases capture functional requirements, define system.

Notas

Tema | Día | Mes | Año | ESTILO

1.05.09

Objects and classes

Objects represent individual things with state and identity while classes are abstractions grouping objects with common properties

Perspective

Elements can be viewed from analysis, specification or implementation perspectives.

Class structure

Classes are divided into compartments showing names, attributes, operations and responsibilities.

Relationships

Classes connect through generalizations, associations with multiplicity constraints, aggregations, compositions and dependencies.

1.05.10

UML QUICK REFERENCE

UML is a software design system utilizing analysis, specification, and implementation perspectives.

It employs use cases for user goals, class diagrams for static structure, activity diagrams for workflows, and interaction and state diagrams for dynamic behavior and object lifecycles

Notas